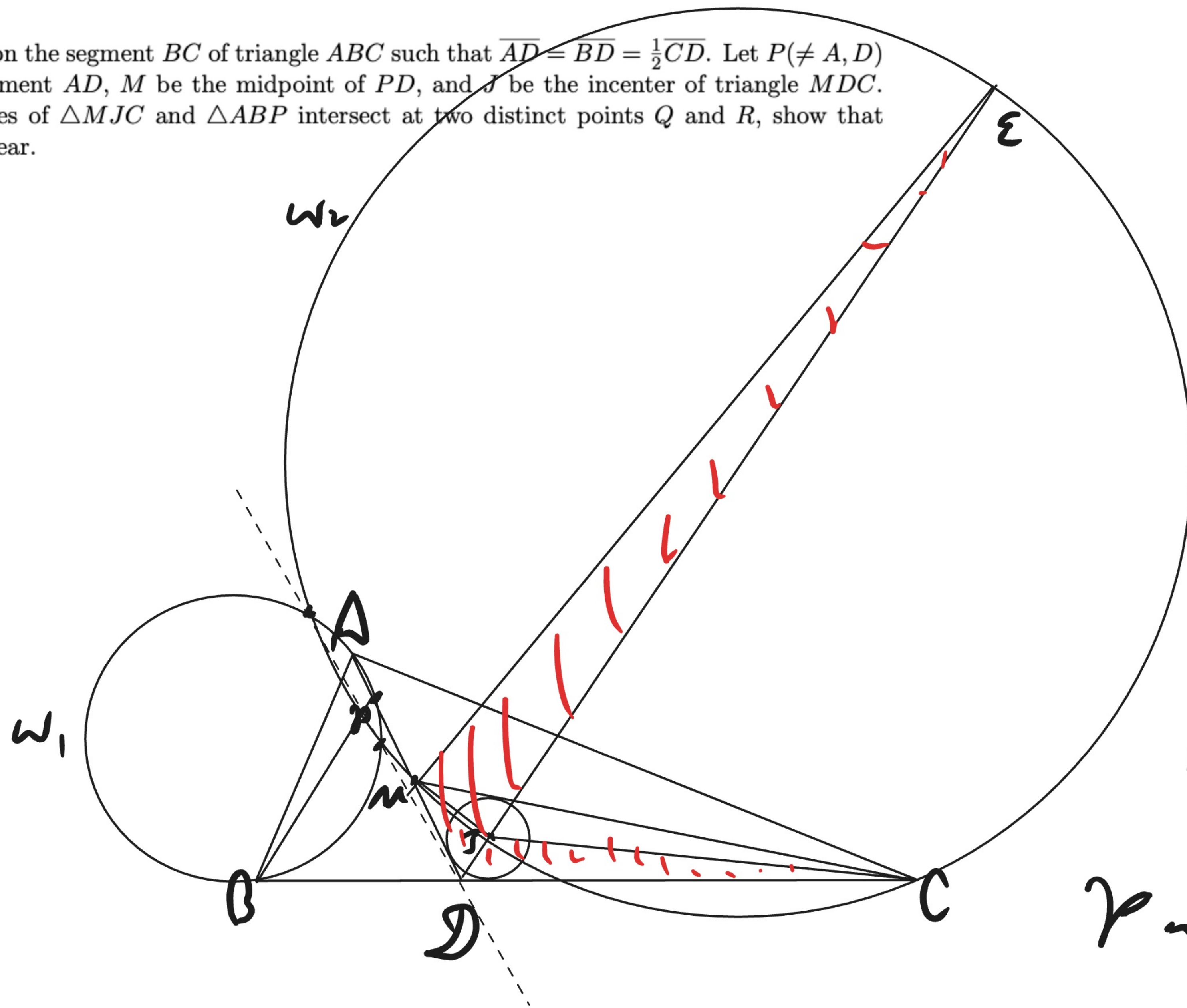


Let D be a point on the segment BC of triangle ABC such that $\overline{AD} = \overline{BD} = \frac{1}{2}\overline{CD}$. Let $P(\neq A, D)$ be a point on segment AD , M be the midpoint of PD , and J be the incenter of triangle MDC . If the circumcircles of $\triangle MJC$ and $\triangle ABP$ intersect at two distinct points Q and R , show that D, Q, R are collinear.



$$QR \Rightarrow \text{collinear?}$$

$$\underline{r_{w_1}(\theta) = r_{w_2}(\theta)?}$$

$$r_{w_1}(\theta) = AD \cdot PD$$

$$= (BD \cdot 2) \cdot MD$$

$$= CD \cdot MD$$

$$r_{w_2}(\theta) = CD \cdot MD?$$

$$r_{w_2}(\theta) = JD \cdot DE$$

$$= CD \cdot MD$$

$$\square$$

$$\angle MDE = \angle JDC$$

$$\angle MDE = \angle MCE = \angle JCD$$

$$\therefore \triangle MDE \sim \triangle JCD$$

$$JD : CD = MD : DE$$

$$CD \cdot MD = JD \cdot DE$$